

Michael (Misha, Mykhailo) Rashkovetskyi

Postdoctoral researcher in astrophysics (cosmology)
Ofc M2031, 191 W Woodruff Ave, Columbus, OH, 43210

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rashkovetskyi.1@osu.edu
<https://rashkovetskyi.im>

Fields of interest

Large-scale structure: galaxy clustering; cosmic microwave background; Hubble tension; nature of dark energy; primordial Universe

Employment

- **The Ohio State University** Columbus, OH, USA
CCAPP (Center for Cosmology and AstroParticle Physics) Fellow 2025 – 2028
- **Harvard College Observatory** Cambridge, MA, USA
Postdoctoral Fellow 2025

Education

- **Harvard University** Cambridge, MA, USA
Ph.D. in Astronomy and Astrophysics 2020 – 2025
M.A. in Astronomy and Astrophysics, in passing 2022
 - Thesis: *Enhancing the analysis of the large-scale structure of the Universe for cutting-edge cosmological surveys with two-point correlation function and beyond* (May 2025)
 - Advisor: Prof. Daniel Eisenstein
 - Center for Astrophysics | Harvard & Smithsonian
- **Tel Aviv University** Tel Aviv-Yafo, Israel
B.Sc. in Physics, Summa Cum Laude (GPA: 98/100) 2019 – 2020
 - Raymond & Beverly Sackler School of Physics & Astronomy
 - Advisor: Dr. Omer Bromberg
- **Moscow Institute of Physics and Technology** Dolgoprudny, Russia
B.Sc. in Applied Mathematics and Physics, unfinished 2015 – 2018
 - Department of General and Applied Physics
 - Advisor: Prof. Vasily Beskin
- **Richelieu Lyceum** Odesa, Ukraine
High school, specialization in physics 2010 – 2015

First-author publications

4. M. **Rashkovetskyi**, D. J. Eisenstein, *et al.*, “Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect”, *The Open Journal of Astrophysics* **8** October (2025)a 46033, [arXiv:2508.20904](https://arxiv.org/abs/2508.20904)

3. M. **Rashkovetskyi**, D. Forero-Sánchez, A. de Mattia, D. J. Eisenstein, N. Padmanabhan, H. Seo, A. J. Ross, *et al.*, “Semi-analytical covariance matrices for two-point correlation function for DESI 2024 data”, *J. Cosmology Astropart. Phys.* **2025** January (2025)b 145, [arXiv:2404.03007](#)
2. M. **Rashkovetskyi**, D. J. Eisenstein, *et al.*, “Validation of semi-analytical, semi-empirical covariance matrices for two-point correlation function for early DESI data”, *MNRAS* **524** September (2023) 3894–3911, [arXiv:2306.06320](#)
1. M. **Rashkovetskyi**, J. B. Muñoz, D. J. Eisenstein, and C. Dvorkin, “Small-scale clumping at recombination and the Hubble tension”, *Phys. Rev. D* **104** November (2021) 103517, [arXiv:2108.02747](#)

First-tier author publications

10. E. A. Zaborowski, P. Taylor, K. Honscheid, A. Cuceu, A. de Mattia, A. Krolewski, M. **Rashkovetskyi**, A. J. Ross, C. To, *et al.*, “ H_0 Without the Sound Horizon (or Supernovae): A 2% Measurement in DESI DR1”, October 2025, [arXiv:2510.19149](#)
9. D. Valcin, M. **Rashkovetskyi**, H. Seo, F. Beutler, P. McDonald, A. de Mattia, A. J. Rosado-Marín, A. J. Ross, N. Padmanabhan, *et al.*, “Combined tracer analysis for DESI 2024 BAO”, August 2025, [arXiv:2508.05467](#)
8. U. Andrade, E. Paillas, J. Mena-Fernández, Q. Li, A. J. Ross, S. Nadathur, M. **Rashkovetskyi**, A. Pérez-Fernández, H. Seo, N. Sanders, O. Alves, X. Chen, N. Deiosso, A. de Mattia, M. White, *et al.*, “Validation of the DESI DR2 measurements of baryon acoustic oscillations from galaxies and quasars”, *Phys. Rev. D* **112** October (2025) 083512, [arXiv:2503.14742](#)
7. D. Forero-Sánchez, M. **Rashkovetskyi**, O. Alves, *et al.*, “Analytical and EZmock covariance validation for the DESI 2024 results”, *J. Cosmology Astropart. Phys.* **2025** April (2025) 055, [arXiv:2411.12027](#)
6. J. Yu, A. J. Ross, A. Rocher, O. Alves, A. de Mattia, D. Forero-Sánchez, J.-P. Kneib, A. Krolewski, T. Lan, M. **Rashkovetskyi**, *et al.*, “ELG spectroscopic systematics analysis of the DESI Data Release 1”, *J. Cosmology Astropart. Phys.* **2025** January (2025) 126, [arXiv:2405.16657](#)
5. J. Mena-Fernández, C. Garcia-Quintero, S. Yuan, B. Hadzhiyska, O. Alves, M. **Rashkovetskyi**, H. Seo, N. Padmanabhan, S. Nadathur, C. Howlett, S. Alam, A. Rocher, A. J. Ross, E. Sanchez, M. Ishak, *et al.*, “HOD-dependent systematics for luminous red galaxies in the DESI 2024 BAO analysis”, *J. Cosmology Astropart. Phys.* **2025** January (2025) 133, [arXiv:2404.03008](#)
4. C. Garcia-Quintero, J. Mena-Fernández, A. Rocher, S. Yuan, B. Hadzhiyska, O. Alves, M. **Rashkovetskyi**, H. Seo, N. Padmanabhan, S. Nadathur, C. Howlett, M. Ishak, L. Medina-Varela, P. McDonald, A. J. Ross, Y. Xie, X. Chen, A. Bera, *et al.*, “HOD-dependent systematics in Emission Line Galaxies for the DESI 2024 BAO analysis”, *J. Cosmology Astropart. Phys.* **2025** January (2025) 132, [arXiv:2404.03009](#)
3. J. Moon, D. Valcin, M. **Rashkovetskyi**, C. Saulder, *et al.*, “First detection of the BAO signal from early DESI data”, *MNRAS* **525** November (2023) 5406–5422, [arXiv:2304.08427](#)
2. E. M. Novoselov, V. S. Beskin, A. K. Galishnikova, M. M. **Rashkovetskyi**, and A. V. Biryukov, “Orthogonal pulsars as a key test for pulsar evolution”, *MNRAS* **494** April (2020) 3899–3911, [arXiv:2004.03211](#)
1. V. S. Beskin, A. K. Galishnikova, E. M. Novoselov, A. A. Philippov, and M. M. **Rashkovetskyi**, “So how do radio pulsars slow-down?”, in “*Journal of Physics Conference Series*”, vol. 932, p. 012012. December 2017

Collaboration publications

19. C. Garcia-Quintero, H. E. Noriega, A. de Mattia, A. Aviles, K. Lodha, D. Chebat, J. Rohlf, S. Nadathur, W. Elbers, *et al.*, “Cosmological implications of DESI DR2 BAO measurements in light of the latest ACT DR6 CMB data”, April 2025, [arXiv:2504.18464](#)
18. DESI Collaboration *et al.*, “DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints”, *Phys. Rev. D* **112** October (2025)a 083515, [arXiv:2503.14738](#)
17. DESI Collaboration *et al.*, “DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest”, *Phys. Rev. D* **112** October (2025)b 083514, [arXiv:2503.14739](#)
16. K. Lodha, R. Calderon, W. L. Matthewson, A. Shafieloo, M. Ishak, J. Pan, C. Garcia-Quintero, D. Huterer, G. Valogiannis, L. A. Ureña-López, N. V. Kamble, D. Parkinson, A. G. Kim, G. B. Zhao, J. L. Cervantes-Cota, J. Rohlf, F. Lozano-Rodríguez, J. O. Román-Herrera, *et al.*, “Extended dark energy analysis using DESI DR2 BAO measurements”, *Phys. Rev. D* **112** October (2025) 083511, [arXiv:2503.14743](#)
15. W. Elbers, A. Aviles, H. E. Noriega, D. Chebat, A. Menegas, C. S. Frenk, C. Garcia-Quintero, D. Gonzalez, M. Ishak, O. Lahav, K. Naidoo, G. Niz, C. Yèche, *et al.*, “Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape”, *Phys. Rev. D* **112** October (2025) 083513, [arXiv:2503.14744](#)
14. DESI Collaboration *et al.*, “Data Release 1 of the Dark Energy Spectroscopic Instrument”, March 2025, [arXiv:2503.14745](#)
13. X. Chen, Z. Ding, E. Paillas, S. Nadathur, H. Seo, S. Chen, N. Padmanabhan, M. White, A. de Mattia, P. McDonald, A. J. Ross, A. Variu, A. Carnero Rosell, B. Hadzhiyska, M. M. S. Hanif, D. Forero-Sánchez, *et al.*, “Extensive analysis of reconstruction algorithms for DESI 2024 baryon acoustic oscillations”, November 2024, [arXiv:2411.19738](#)
12. DESI Collaboration *et al.*, “DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics”, *J. Cosmology Astropart. Phys.* **2025** July (2025)a 017, [arXiv:2411.12020](#)
11. DESI Collaboration *et al.*, “DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars”, *J. Cosmology Astropart. Phys.* **2025** September (2025)b 008, [arXiv:2411.12021](#)
10. DESI Collaboration *et al.*, “DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements”, *J. Cosmology Astropart. Phys.* **2025** July (2025)c 028, [arXiv:2411.12022](#)
9. A. Pérez-Fernández, L. Medina-Varela, R. Ruggeri, M. Vargas-Magaña, H. Seo, N. Padmanabhan, M. Ishak, *et al.*, “Fiducial-cosmology-dependent systematics for the DESI 2024 BAO analysis”, *J. Cosmology Astropart. Phys.* **2025** January (2025) 144, [arXiv:2406.06085](#)
8. U. Andrade, J. Mena-Fernández, H. Awan, A. J. Ross, S. Brieden, J. Pan, A. de Mattia, *et al.*, “Validating the galaxy and quasar catalog-level blinding scheme for the DESI 2024 analysis”, *J. Cosmology Astropart. Phys.* **2025** January (2025) 128, [arXiv:2404.07282](#)
7. DESI Collaboration *et al.*, “DESI 2024 III: baryon acoustic oscillations from galaxies and quasars”, *J. Cosmology Astropart. Phys.* **2025** April (2025)a 012, [arXiv:2404.03000](#)
6. DESI Collaboration *et al.*, “DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest”, *J. Cosmology Astropart. Phys.* **2025** January (2025)b 124, [arXiv:2404.03001](#)

5. DESI Collaboration *et al.*, “DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations”, J. Cosmology Astropart. Phys. **2025** February (2025)c 021, [arXiv:2404.03002](#)
4. E. Paillas, Z. Ding, X. Chen, H. Seo, N. Padmanabhan, A. de Mattia, A. J. Ross, S. Nadathur, C. Howlett, *et al.*, “Optimal reconstruction of baryon acoustic oscillations for DESI 2024”, J. Cosmology Astropart. Phys. **2025** January (2025) 142, [arXiv:2404.03005](#)
3. S. F. Chen, C. Howlett, M. White, P. McDonald, A. J. Ross, H. J. Seo, N. Padmanabhan, *et al.*, “Baryon acoustic oscillation theory and modelling systematics for the DESI 2024 results”, MNRAS **534** October (2024) 544–574, [arXiv:2402.14070](#)
2. DESI Collaboration *et al.*, “Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument”, AJ **167** February (2024)a 62, [arXiv:2306.06307](#)
1. DESI Collaboration *et al.*, “The Early Data Release of the Dark Energy Spectroscopic Instrument”, AJ **168** August (2024)b 58, [arXiv:2306.06308](#)

Outreach and teaching

- **DESI redshift-space distortions animations** with Claire Lamman+
Early version used in the press-release and following news articles Nov 19, 2024 – Mar 2, 2025
- **Astronomy 200: Radiative Processes in Astrophysics** Harvard University
Teaching Fellow Fall 2023
- **Astronomy 201: Astrophysical Fluids & Plasmas** Harvard University
Teaching Fellow Spring 2023
- **Astronomy 130: Introduction to Cosmology** Harvard College
Teaching Fellow Fall 2022

Seminar talks

- **Ohio University Astro Seminar** Athens, OH, USA
Enhancing the analysis of the large-scale structure of the Universe October 8, 2025
- **CCAPP Fellows Symposium 2025** Columbus, OH, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect September 17, 2025
- **KIPAC Tea (Stanford / SLAC)** Menlo Park, CA, USA
Fast semi-analytical covariance matrices for 2-point correlation functions May 10, 2024
- **ITC Luncheon** Cambridge, MA, USA
Semi-analytic covariance matrices for 2PCF of DESI galaxies (on YouTube) September 28, 2023

Contributed conference talks

- **APS Global Physics Summit** Denver, CO, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect March 17, 2026
- **28th International Conference on Particle Physics & Cosmology** Pittsburgh, PA, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect October 16, 2025

- **APS Global Physics Summit** Anaheim, CA, USA
Clustering of DESI LRG selected based on ACT DR6 + Planck tSZ map March 18, 2025
- **IX Essential Cosmology for the Next Generation** Playa del Carmen, Mexico
Clustering of DESI galaxies selected based on ACT thermal SZ map December 5, 2024
- **Cosmology from Home 2023** online
First Detection of the BAO Signal from Early DESI Data (on YouTube) July 4, 2023
Co-presenter with Christoph Saulder, Jeongin Moon and David Valcin
- **APS April Meeting** Minneapolis, MN, USA
First Detection of the BAO Signal from Early DESI Data April 18, 2023

Poster presentations

- **Arizona Winter School on Multi-Probe Cosmology** Tucson, AZ, USA
Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect January 12-15, 2025
Second best poster prize
- **Frontiers in Cosmology and Gravitational Physics** Portsmouth, UK
Fast semi-analytical covariance matrices for 2PCF of galaxies and quasars May 20-23, 2024
- **VIII Essential Cosmology for the Next Generation** Playa del Carmen, Mexico
RascalC: Empirical 2PCF Covariance Matrices without Mocks Nov 30 - Dec 3, 2022
- **CMB-S4 Summer Meeting** online
Small-scale Clumping at Recombination and the Hubble Tension August 9-13, 2021
- **CMB-S4 Spring Meeting** online
Hubble Tension with Small-Scale Clumping March 8-12, 2021
- **Physics of Neutron Stars - 2017** Saint-Petersburg, Russia
On the light-curve anomalies of radio pulsars July 10-14, 2017

Other meetings

- **DESI December** Tucson, AZ, USA
DESI Collaboration December 8-11, 2025
- **DESI July** Berkeley, CA, USA
DESI Collaboration July 8-11, 2025
- **DESI December** Cancún, Mexico
DESI Collaboration December 10-13, 2024
- **DESI July** Marseille, France
DESI Collaboration July 9-12, 2024
- **Fundamental Physics from Future Spectroscopic Surveys** Berkeley, CA, USA
Lawrence Berkeley National Laboratory May 6-8, 2024
- **DESI December** Waikoloa, HI, USA
DESI Collaboration December 11-14, 2023
- **DESI July** Durham, UK
DESI Collaboration July 17-21, 2023
- **2nd Michigan Cosmology Summer School** Ann Arbor, MI, USA
University of Michigan June 5-9, 2023

- **DESI December** Cancún, Mexico
DESI Collaboration December 5-9, 2022
- **DESI June** Berkeley, CA, USA
DESI Collaboration June 21-24, 2022
- **15th School of Modern Astrophysics** Dolgoprudny, Russia
Moscow Insitute of Physics and Technology July 1-12, 2019
- **13th School of Modern Astrophysics** Dolgoprudny, Russia
Moscow Insitute of Physics and Technology July 3-21, 2017
- **International School of Subnuclear Physics - 2017** Erice, Italy
"Ettore Majorana" Foundation and Centre for Scientific Culture June 14-23, 2017
- **International school on particles, fields and strings** Moscow, Russia
National Research University "High School of Economics" April 17-24, 2017
- **Astronomical practice** Nizhniy Arkhyz, Russia
Special Astrophysical Observatory June 25 – July 2, 2016

Awards, grants and honors

Dean's Certificate in Recognition of Outstanding Academic Achievements (TAU) 2019–2020
 Stipend for excellent students of MIPT in the name of A.Abramov 2016–2017
 International Physics Olympiad, bronze medal Mumbai, 2015
 International Physics Olympiad, silver medal Astana, 2014

Selected open source contributions (<https://github.com/misharash>)

- RASCALC C++ & Python
Fast semi-analytic covariance matrices library/program 2022-2024
- RASCALC scripts Python
DESI covariance matrix pipeline for 2-point correlation function (scripts) 2024
- PYCORR Python
Library for 2-point correlation function estimation 2024

Languages

- Russian: native
- Ukrainian: fluent
- English: advanced
- Hebrew: advanced
- German: intermediate